

Introducing Hyperledger Quilt, a Java Implementation of the Interledger Protocol

Hyperledger Quilt is a project hosted by the Linux Foundation. It offers interoperability between ledger systems by implementing the Interledger Protocol (ILP). It is designed to be a secure payment tool for enterprises while isolating senders and receivers from the risks of intermediary failures.



Payment networks today operate in silos and hence are disconnected. Payments are relatively easy within one country or if the sender and recipient have accounts in the same network or ledger. However, sending from one ledger to another is often impossible. Where connections do exist, they are manual, slow, or expensive.

The Interledger Protocol

Also known as ILP, this is a protocol for making transactions across ledgers. The protocol's standards and specifications are being defined by the open source community under the umbrella of the World Wide Web Consortium. ILP provides for routing payments across different digital asset ledgers while isolating senders and receivers from the risks of intermediary failures. Secure multi-hop payments and automatic routing enables a global network of networks for making payments of different values, connecting any sender with any receiver.

The Interledger Protocol is intentionally limited in scope to provide the functions necessary to deliver a payment from a source to a destination over an interconnected system of

ledgers. It includes minimal requirements from the underlying ledgers, and it does not include public key infrastructure, identity and liquidity management or other services commonly found in payment protocols.

The central functions of the ILP are addressing hosts and securing payments across different ledgers. Each host that sends and receives interledger payments has an interledger module that uses the addresses in the interledger header to transmit interledger payments towards their destinations. Interledger modules share common rules for interpreting addresses. The modules also have procedures for making routing decisions and other functions.

The ILP uses transfer holds to ensure that the sender's funds are either delivered to the destination account or returned to the sender's account. This mechanism is described in greater detail in the Interledger whitepaper [<https://interledger.org/interledger.pdf>].

The Interledger protocol is based on concepts dating back to the 1970s and 1980s, but it took the advent of Bitcoin and the global blockchain movement to make the world